

James Lewis - S-29

Experiment 08

Code:

```
import threading
import time

def print_numbers():
    for i in range(1, 6):
        print(f"Thread 1 - Number: {i}")
        time.sleep(1)

def print_letters():
    for letter in 'ABCDE':
        print(f"Thread 2 - Letter: {letter}")
        time.sleep(1)

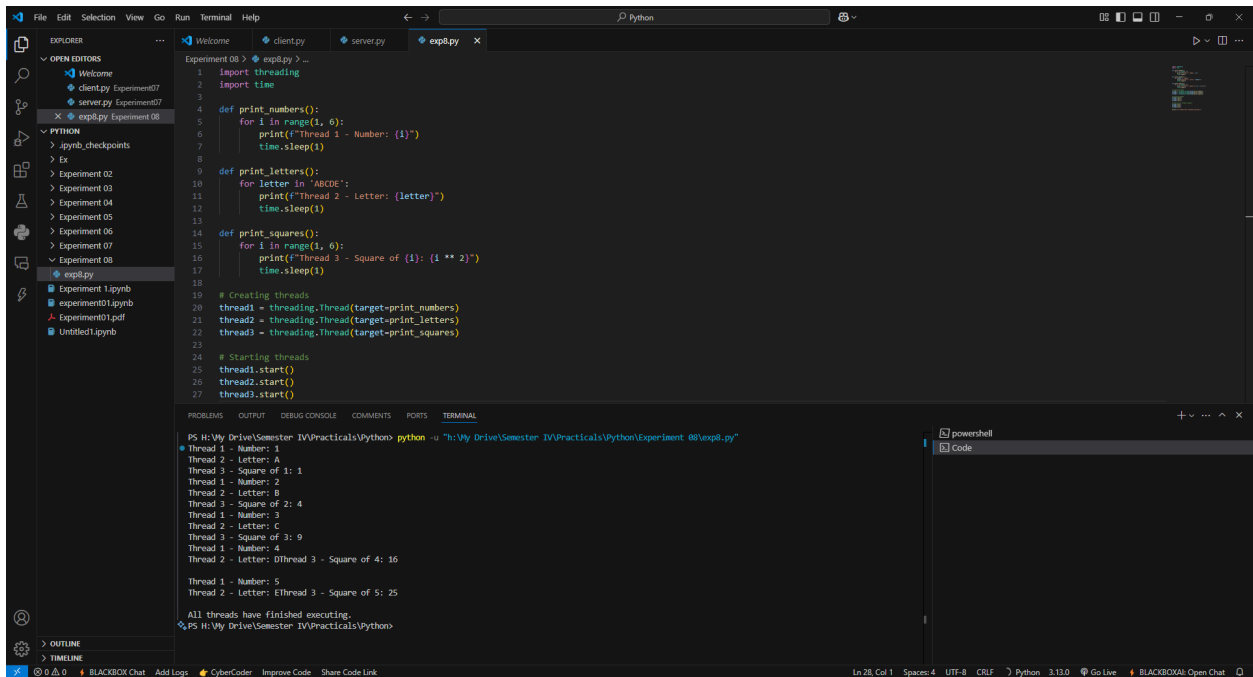
def print_squares():
    for i in range(1, 6):
        print(f"Thread 3 - Square of {i}: {i ** 2}")
        time.sleep(1)

# Creating threads
thread1 = threading.Thread(target=print_numbers)
thread2 = threading.Thread(target=print_letters)
thread3 = threading.Thread(target=print_squares)

# Starting threads
thread1.start()
thread2.start()
thread3.start()

# Ensuring all threads complete
thread1.join()
thread2.join()
thread3.join()
print("All threads have finished executing.")
```

Implement



```
1 import threading
2 import time
3
4 def print_numbers():
5     for i in range(1, 6):
6         print(f"Thread 1 - Number: {i}")
7         time.sleep(1)
8
9 def print_letters():
10     for letter in 'ABCDE':
11         print(f"Thread 2 - Letter: {letter}")
12         time.sleep(1)
13
14 def print_squares():
15     for i in range(1, 6):
16         print(f"Thread 3 - Square of {i}: {i ** 2}")
17         time.sleep(1)
18
19 # Creating threads
20 thread1 = threading.Thread(target=print_numbers)
21 thread2 = threading.Thread(target=print_letters)
22 thread3 = threading.Thread(target=print_squares)
23
24 # Starting threads
25 thread1.start()
26 thread2.start()
27 thread3.start()
28
29 # All threads have finished executing.
```

PS H:\My Drive\Semester IV\Practicals\Python> python -u "h:\My Drive\Semester IV\Practicals\Python\Experiment 08\exp8.py"

```
Thread 1 - Number: 1
Thread 2 - Letter: A
Thread 3 - Square of 1: 1
Thread 1 - Number: 2
Thread 2 - Letter: B
Thread 3 - Square of 2: 4
Thread 1 - Number: 3
Thread 2 - Letter: C
Thread 3 - Square of 3: 9
Thread 1 - Number: 4
Thread 2 - Letter: D
Thread 3 - Square of 4: 16
Thread 1 - Number: 5
Thread 2 - Letter: E
Thread 3 - Square of 5: 25
All threads have finished executing.
```

Output:

```
PS H:\My Drive\Semester IV\Practicals> python -u "h:\My Drive\Semester IV\Practicals\Python\Experiment 08\exp8.py"

Thread 1 - Number: 1
Thread 2 - Letter: A
Thread 3 - Square of 1: 1
Thread 1 - Number: 2
Thread 2 - Letter: B
Thread 3 - Square of 2: 4
Thread 1 - Number: 3
Thread 2 - Letter: C
Thread 3 - Square of 3: 9
Thread 1 - Number: 4
Thread 2 - Letter: D
Thread 3 - Square of 4: 16
Thread 1 - Number: 5
Thread 2 - Letter: E
Thread 3 - Square of 5: 25
All threads have finished executing.
```